

Problem Solving in Engineering (PSE): Student Workbook

Systems Thinking for Problem Solving

AUTHOR: SAMPADA PACHAURY

Workbook Overview & Guidelines

This workbook is a structured guide for students to follow as part of **Problem Solving in Engineering (PSE)** – IP-3 Experiential Learning Course

Follow the modules outlined in this workbook, you will adopt and practice systems thinking methodologies to solve real-world problems in a systemic manner.

Until now, you spotted a problem or a gap, you thought of an idea and you developed a prototype. The downside of this rushed problem solving is that the root cause of the problem is not addressed and hence the problem can recur or newer problems can appear. Metaphor recall – 10-headed snake.

Through this course we want you to choose a system to fix instead of a problem.

You will learn and apply Systems Thinking methods to formulate the problem and develop resilient, robust and sustainable solutions.

How to Use This Workbook

- **Collaborative Teams:** Work in mixed-disciplinary teams of 3-5 team members.
- **Active Writing:** Use the fillable workspace to document your team's discussions, data, models, design and new system design architecture.
- **Technology Agnostic Philosophy:** The work book is generic and not bound to a disciplinary domain. It will guide you with methods and tools. You are expected to build your system awareness and expand your understanding by referring to links at the end of each module and apply them to your specific case.
- **Iterative Progression:** You will begin with Module 1, but the progress is not intended to be linear from module 1 to 6. It is recommended that you move back and forth from Module 1 to 4 as per your research and investigation. The best results will come if you do not remain stuck in a module. Keep the momentum alive and progress to subsequent modules while keeping an option open to come back and work on previous modules.
- **Process over Product:** In your first iteration of the workbook, you will develop the abilities to think in systems. Your solution may not be the best fix. The goal is to learn how to adopt systems thinking and rapid prototyping, not to create a flawless product. Continue the journey of PSE and you will be able to propose better system fixes in your next round of iterations. You can start with a fresh workbook.

Gen AI Usage Guidelines

- **Establish context:** Introduce yourself with your role in the project, project objectives and expectations and limitations that warrant use of AI
- **AI Roleplay:** Define what role will AI play in this partnership, e.g. Assistant/Critique/Writer/Advisor/Peer-Reviewer/Subject Matter Expert of []
- **Set Scope:** Specify AI deliverables and expectations clearly, some examples;
 - Cite sources
 - Exclude opinions not backed by strong evidence
 - Restrict all references to count of [] citations or published journals
 - Support information with reasoning

- Provide data but do not analyse and draw conclusions
- Skip ice-breaking sentences and do not bring social human-like flavour to the responses
- Provide contradicting perspectives as an unbiased observer
- Question my assumptions and ask for my reasons to my claims

Affected System (Problem Area) Selection

Identify a problem area or affected system that you would like to take up as your project in PSE (IP-3).

Select a Super System

For example: Transportation & Sustainable Mobility System

Select a Specific Sub System (Problem Area) to focus on:

For example: TGSRTC – Telangana State Road Transport System –

Specific Location: Hyderabad.

Specific Issue – Protests

Specific Focus – Driver and Conductor

Problem Area if identified (Describe the local pain point in 2-3 sentences):

For example: TGSRTC Driver and Conductor Health Condition - TGSRTC has faced protests over long working shifts (12-15 hours) of the drivers and conductors and the impact on their physical and mental health.

Module 1: Drafting the Project Informatics Document (PID)

Scoping Real-World Challenges and Building the Project Informatics Document

Module Targets/Outcome

- Scope a specific, localized problem within India (for e.g., Hyderabad IT corridor, rural Telangana).
- Formulate a **Project Informatics Document (PID)** establishing system boundaries, stakeholders, and physical interfaces/data baselines.

Study the System where the problem is observed

Describe what is happening in the System. Use **5W2H suggested below** to guide your inquiry.

Please Note: This is an As-Is State representation and should NOT be mixed with the “should be” state.

Collect all the background data about the problem area from official sources. (e.g., citing reports from NITI Aayog, MoHUA, or local municipal bodies), web based links, news articles, research papers, reviews, stakeholder interviews etc.

The data or information is inadmissible if the sources are not cited

All the claims must be backed with evidence. No evidence, No Claim

At the end, your PID must have a statistical base and facts and figures.

Keep in mind:

- Are the sources official, recognised and citable?
- Is every claim backed by facts and figures?
- Is there enough measurable/quantified content to draw conclusions?
- If there are multiple sources citing conflicting claims, which ones would you refer to and what conclusions would you draw from the conflict?
- **What are the associated assumptions?**
- **What variables influence the observed claims? What changes when the variables change and is there enough evidence to this?**

Important Note: Keep your choice of technology area out of the context at the time of drafting PID. The focus is on the Problem area and evolving a deeper understanding of the system in which the problem exists. Do NOT investigate with a pre-conditioned mindset that you know the problem. Capture Symptoms and the context around the symptoms but do not associate it with an assumption which cannot be validated with data. You do NOT know the real problem yet.

5W2H Strategy relevant for PID

WHAT is/are the SYMPTOMS (**Effect**) of the problem observed?

WHO is affected most by the problem? (Stakeholder group 1) (List all)

WHO is **Causing** the Problem? (List all if applicable) (Stakeholder Group 2)

WHEN (Time patterns) is the Symptom happening? (Should contrast with the times when the symptoms are not occurring)

WHERE (Space context) is/are the symptoms observed? (Should eliminate the locations where the problem does not occur)

HOW is the process cycle observed? (How does the information or data move from one state to another?) (Draw a life cycle diagram or a flow chart)

HOW MUCH is the impact? Each claim above must be backed by the data/statistics. Provide count and cost.

Instructions:

Use this space to record all relevant observations, facts, statistics, stakeholder inputs, reports, research findings, and other supporting information related to the selected problem area. Focus on collecting information rather than analysing it or proposing solutions. Additional sheets may be attached if more space is required.

Write the Problem Statement (preliminary)

State the Problem in less than 100 words. This is your current understanding of the problems observed in the System. In case of multiple problems observed, mention as many as you have observed.

- Are these problems related to each other?
- What is the Causal relationship between the problems?
- Which one problem do you observe to be the cause of other problems if they are found related?
- Is there an obvious vicious loop (feedback loop) that causes the problems to compound?

Define the System

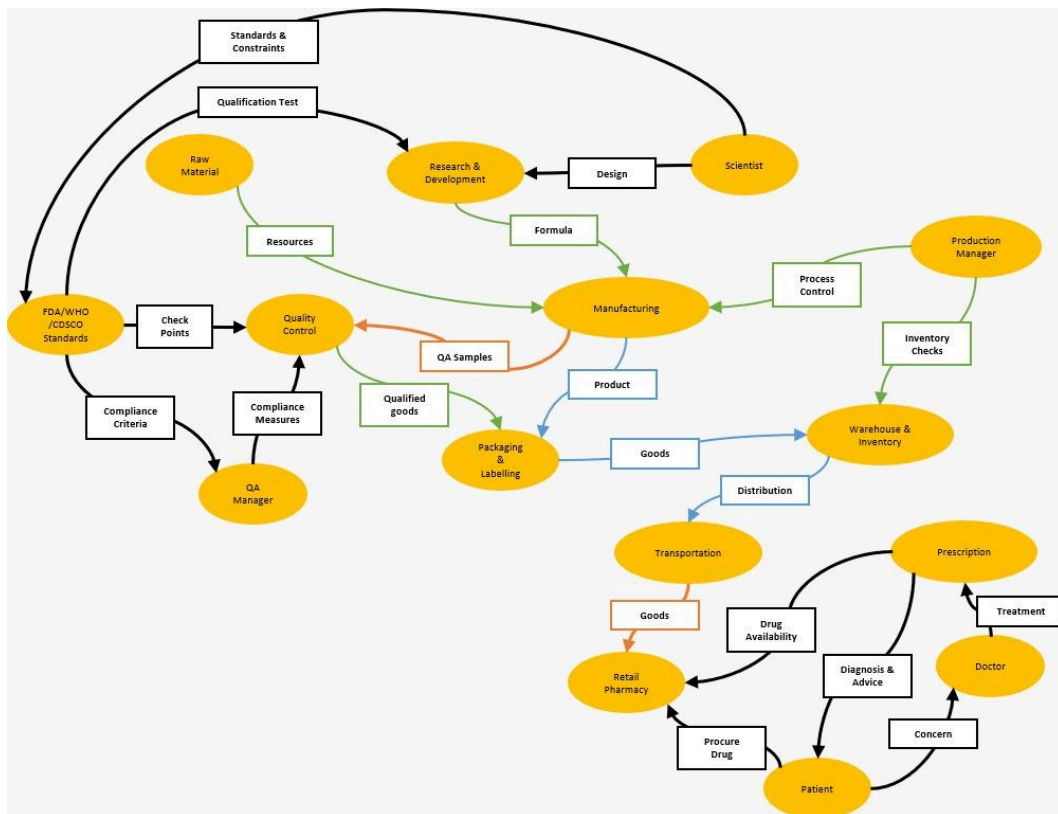
List all the components of the System. Each component is defined by its function.

For example; a part could be the role a person plays and not the name of the person itself. A Part could be a Smartphone User Interface, Installed Applications, Camera, Phone User, Retail shop, Price, Model etc.

After you have listed all components that you can think of, connect components to each other to indicate the relationships between the components and form a System map. The connectors could be one-way or two-way arrows suggesting how the components interact with each other. For better understanding the relationship can be written on the connector.

Watch the Video “Understanding Systems” on Youtube with link provided at the end of the module

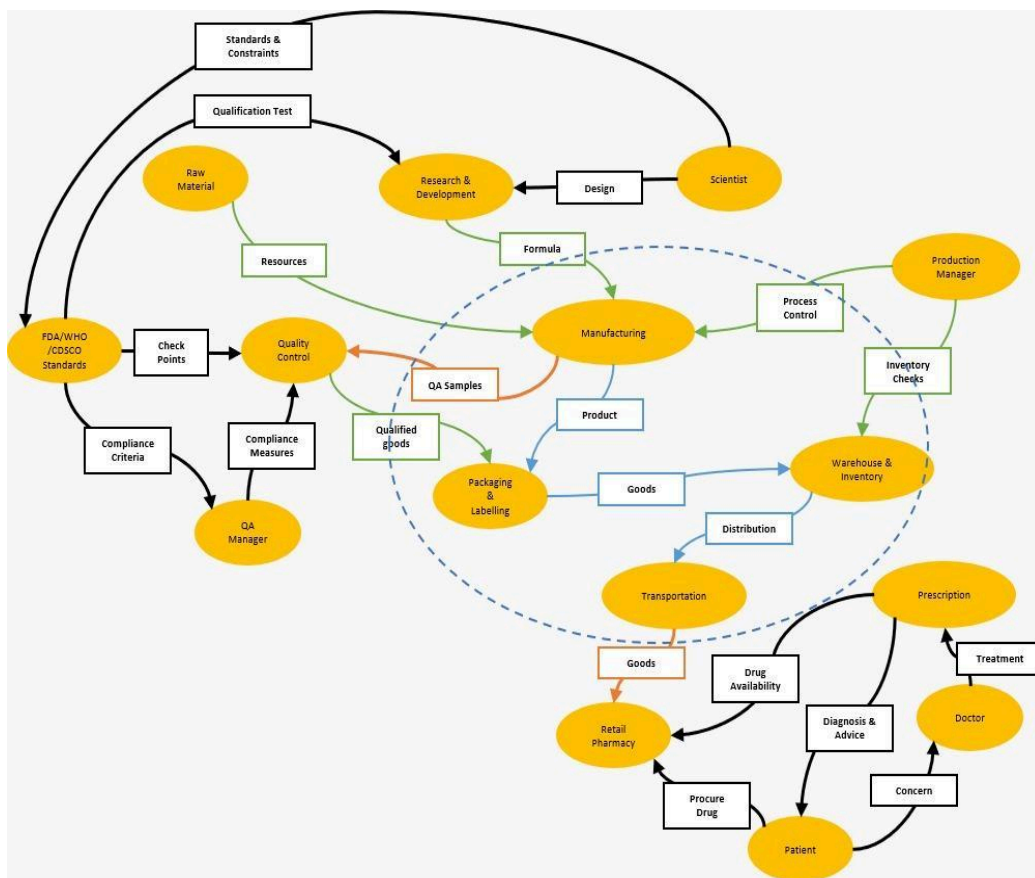
Example: System Map of a Drug Manufacturing supply chain.



Define the System Boundary (Scope)

The system boundary defines what is included within the scope of your study and what remains outside it (the environment). The inputs drive the function of the system. The output from the system can be classified as desirable and not-so-desirable (waste) output.

Example given: System Map of a Drug Manufacturing Supply Chain System where the problem area focus is on supply chain from manufacturing to retail encircled within the system boundary depicted as dashed.



*Inputs are identified as **green** arrows entering into the scope of your system from the environment and outputs are identified as **red** arrows leaving the environment.*

*Note: The dotted boundary will be the scope of the System which you will be working on.
(The scope of your system can be modified later to change the sub systems you intend to work upon.)*

Stakeholder Map

Primary Stakeholder Map: List at least three distinct stakeholder groups and their primary needs.

For Example:

Stakeholder: Paddy Farmers

Need: Android mobile camera interface, localized voice-over output [Cite Source].

Note: If your project involves more than three stakeholder groups, you can always add additional entries.

1 Stakeholder:

Needs:

2 Stakeholder:

Needs:

3 Stakeholder:

Needs:

Quantitative Baseline Data: State at least two localized statistics or data points relevant to your problem from your PID. Cite sources.

For example:

*Data Point: Mobile internet penetration in rural Telangana has crossed **68%**, but network bandwidth in fields remains highly unstable (averaging **less than 1.5 Mbps** upload speeds in interior fields), making heavy cloud-based AI models completely unviable for real-time field diagnosis [Cite Source]*

- 1 Data Point A: *e.g. Average daily traffic volume at the selected intersection is **18,500 vehicles per day**, with peak-hour congestion occurring between **8:00 AM and 10:00 AM**.*

Source:

- 2 Data Point B: *for example the average waiting time at the traffic signal during peak hours is **4.8 minutes**, leading to increased fuel consumption and delays.*

Source:

State “How Might We...”

Write three "How Might We" (HMW) statements that define the transition from the existing system to the desired system. Each statement should highlight the problem your project addresses and the improvement you want to achieve.

How might we move from **(existing system or challenge)** to a system that **(achieves the desired outcome)**?

For example: How might we compress a deep-learning crop disease classification model to run efficiently on low-cost Android smartphones with limited RAM and processing power?

1 How Might We

2 How Might We

3 How Might We

Module 1 Resources & Reading Links/Video Links

- Thinking in Systems – Donella Meadows
- The Waters Center for Systems Thinking: **waterscenterst.org**
- Understanding Systems: <https://www.youtube.com/watch?v=DVoy02xd2Mg>

Video Chapters:

- Complexity ([00:00](#))
- Hidden Patterns ([02:17](#))
- Wicked Problems ([06:37](#))
- Understanding the System ([09:28](#))
- Systems Thinking ([17:37](#))
- Zoom In & Zoom Out ([27:48](#))
- Long-Term Impact ([28:50](#))
- Feedback Loops ([33:01](#))
- Interdisciplinary Collaboration ([34:52](#))
- Applying Systems Thinking ([35:44](#))
- Modeling (Trends) ([36:58](#))
- Leverage Points ([39:33](#))